

EMPORIA-GREENSVILLE, VA, USA

WMO: 720278

Lat: 36.687N

Lon: 77.483W

Elev: 39

StdP: 100.86

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.9	-17.4	0.8	-4.3	-14.8	1.0	-2.8	8.5	8.8	7.6	8.5	1.2	290	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	34.4	25.0	32.7	24.6	32.0	24.3	27.8	30.4	26.1	30.4	25.2	30.0	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	29.7	25.1	20.3	28.1	23.7	18.7	27.3	88.5	30.9	81.4	29.8	76.8	29.7	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.3	5.9	5.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-12.8	36.7	4.2	1.9	-15.8	38.0	-18.3	39.1	-20.7	40.2	-23.7	41.6
			-13.5	28.2	4.0	2.3	-16.4	29.8	-18.7	31.2	-21.0	32.5	-23.9	34.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	4.1	5.4	9.0	14.7	19.1	23.5	24.7	21.7	15.4	9.3	6.3	(6)	
(7)		DBStd	8.76	5.70	5.35	5.46	4.67	4.10	3.04	2.38	2.60	3.17	4.61	4.57	5.09	(7)
(8)		HDD10.0	658	198	148	87	11	0	0	0	0	7	66	140	(8)	
(9)		HDD18.3	2038	442	362	292	128	43	2	0	1	8	115	271	374	(9)
(10)		CDD10.0	2462	14	20	57	152	282	404	481	456	352	174	47	24	(10)
(11)		CDD18.3	799	0	1	3	18	67	157	222	198	110	23	1	1	(11)
(12)		CDH23.3	7311	0	4	41	227	619	1486	2110	1758	852	200	9	3	(12)
(13)		CDH26.7	2831	0	0	4	49	183	591	931	737	291	46	0	0	(13)
(14)	Wind	WSAvg	1.9	2.0	2.2	2.4	2.4	1.9	1.6	1.5	1.3	1.6	1.7	1.8	1.7	(14)
(15)	Precipitation	PrecAvg	1163	86	67	108	82	92	110	129	118	122	84	76	88	(15)
(16)		PrecMax	1549	162	146	187	171	178	230	201	275	506	155	201	176	(16)
(17)		PrecMin	868	28	20	19	30	34	49	62	38	11	0	12	26	(17)
(18)		PrecStd	174	36	31	42	34	35	50	42	59	103	43	47	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.7	27.2	30.2	32.7	36.1	36.3	36.2	33.0	31.3	25.0	22.6	(19)
(20)		MCWB	16.9	17.8	18.2	19.7	23.4	25.1	25.4	26.1	23.6	22.2	18.0	18.9	(20)	
(21)		2%	DB	19.0	21.1	24.1	27.8	30.3	32.8	34.1	33.7	32.0	27.4	22.2	20.1	(21)
(22)		MCWB	15.6	16.0	16.0	18.9	22.5	25.1	25.5	24.9	22.9	21.3	17.5	17.6	(22)	
(23)		5%	DB	17.0	17.8	22.0	26.2	28.7	32.0	32.7	32.3	30.1	25.9	20.0	17.5	(23)
(24)		MCWB	13.4	13.9	15.2	18.1	22.0	24.7	24.9	24.4	22.5	20.0	16.2	15.2	(24)	
(25)		10%	DB	13.6	15.2	19.0	23.6	27.2	30.2	31.4	31.0	27.8	22.9	17.5	15.2	(25)
(26)		MCWB	10.7	10.7	14.0	17.2	20.7	23.6	24.4	23.9	21.9	19.3	14.0	12.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	18.9	20.5	24.7	27.3	30.9	29.6	30.1	26.6	24.3	20.6	20.6	(27)
(28)		MDCB	20.8	21.9	22.8	27.2	28.4	32.0	32.6	33.0	29.0	27.8	22.1	21.2	(28)	
(29)		2%	WB	16.7	17.1	17.9	21.2	23.7	27.1	27.5	26.9	24.9	23.0	18.5	17.6	(29)
(30)		MDCB	18.6	19.4	21.7	24.5	28.1	29.3	30.5	30.6	28.9	25.9	21.3	19.0	(30)	
(31)		5%	WB	13.6	15.0	16.6	19.6	22.5	25.1	26.0	25.7	24.0	21.9	16.9	15.8	(31)
(32)		MDCB	16.0	17.9	20.2	24.3	26.9	30.4	30.9	30.1	27.8	24.5	18.8	17.3	(32)	
(33)		10%	WB	11.0	11.5	14.9	18.0	21.5	24.1	25.1	24.8	23.2	20.2	15.1	13.0	(33)
(34)		MDCB	13.2	14.1	18.3	22.0	25.4	29.3	30.3	29.1	26.1	22.6	17.7	14.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.7	11.6	12.8	13.5	12.4	11.6	11.2	10.9	11.1	12.4	12.3	10.8	(35)	
(36)		5% DB	MCDBR	14.7	15.3	16.2	15.8	13.9	13.0	12.8	12.8	13.4	14.0	14.2	12.7	(36)
(37)		MCWBR	11.1	10.9	9.3	8.4	6.7	6.3	5.7	5.2	5.8	8.1	9.5	10.2	(37)	
(38)		5% WB	MCDBR	13.3	14.1	13.7	12.8	12.2	11.9	11.5	10.9	10.4	11.7	11.3	11.6	(38)
(39)		MCWBR	11.3	11.5	9.7	8.7	7.2	7.1	6.1	5.4	5.6	7.6	8.7	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.316	0.347	0.390	0.436	0.487	0.514	0.512	0.404	0.357	0.321	0.304	(40)	
(41)		taud	2.529	2.487	2.432	2.321	2.219	2.125	2.069	2.059	2.357	2.477	2.534	2.560	(41)	
(42)		Ebn at Noon	898	924	920	892	852	806	780	771	849	864	867	872	(42)	
(43)		Edh at Noon	78	92	107	126	142	156	164	162	114	92	78	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.43	3.15	4.14	5.26	5.89	6.36	6.03	5.46	4.46	3.55	2.69	2.13	(44)	
(45)		RadStd	0.19	0.37	0.33	0.41	0.47	0.41	0.29	0.30	0.40	0.44	0.22	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (65 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+160	+108	(47)

Nomenclature: See separate page